

PrimeSim™ HSPICE® Tips and Tricks Webisode Series

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Chapter 1. PrimeSim™ HSPICE® Tips and Tricks Webisode Series

The PrimeSim HSPICE Tips and Tricks Webisode Series is a collection of publicly accessible, on-demand mini webinars hosted by Synopsys technical experts. They can help make your circuit simulation tasks easier and more productive. They offer technical tips, tricks, and best practices for using PrimeSim HSPICE features and technologies.

Each webisode, about ten minutes long, is focused on a specific technical topic. Its embedded demo case, wherever applicable, is freely available upon request so that you can learn through hands-on experience and reproduce the results shown in the webisode. Existing webisodes cover topics such as time-domain simulation with S-parameter models, IBIS-AMI behavioral models, statistical eye simulation, Monte Carlo simulation and data mining, dynamic and hierarchical accuracy/performance simulation control, MOS transistor reliability modeling and simulation, and small-signal analysis.

The webisode series is regularly expanded with a new webisode added every month on an average.

Ways to Locate and Access the Webisode Series

1. Search online for "HSPICE Webinar" or "HSPICE Webisode", and click on the top search result.
2. Open the landing Web page: <https://www.synopsys.com/verification/ams-verification/hspice/hspice-tips-tricks.html>.

Published Webisodes in the Series

- [Webisode 14: Probe Analog Characteristics of Complex Transistors—Using Subcircuit-based OPs and Output Templates](#)

All transistors are complex, but some transistors are more complex than others. These transistors are so complex that they have to be represented in subcircuits. How would you probe their analog characteristics, such as transconductance, threshold voltage, and drain saturation margin? In this webisode you will learn how you can do just that with your design, without any extra testbenches.

The complete demo case used in the webisode is available for free. You can request your own copy and try it out!

- [Webisode 13: How to Create a MOS Reliability Model Card—An Excel-based Utility for MOS Reliability Analysis Level 3 Model](#)

This webisode explains how you can transform a pile of transistor reliability data or a paper model from the foundry into a SPICE model, in just a few clicks, and simulate their long-term effect on your circuit. The Excel-based utility and the complete demo case used in the webisode are available for free.

- [Webisode 12: Drift-free Transient Simulation for Signal and Power Integrity Analysis—Using deCap-aware Rational Functions](#)

When verifying your designs for signal and power integrity, you may have observed an artificial DC drift in simulated waveforms. This webisode explains the root cause of such a simulation artifact and recommends best practices for its prevention. You will learn how an enhancement on the scattering parameter handling ensures DC drift-free transient simulation in practical designs. An 8GBps differential PRBS simulation is used as an example in the demonstration.

- [Webisode 11: Spurious Phase Noise Analysis—Predicting Reference Spurs in VCO Phase Noise](#)

If you have been struggling to simulate, visualize, and analyze spurious phase noise in PLLs/VCOs, the struggle is over! Watch this webisode to learn how you can easily predict and visualize reference spurs, along with effects from random noise sources, all in a single phase noise simulation run. You will also learn to use the noise table to efficiently analyze noise contributions both from circuit components and spurs. A 45-nm VCO is used in the demonstration.

- [Webisode 10: Oscillator Power Supply Noise Analysis - Fast and Accurate Simulation Using Harmonic Balance Methods](#)

Noise from the power supply can be a significant source of timing jitter in voltage-controlled oscillators (VCOs). In this webisode, you will learn the principles of harmonic balance and periodic AC analysis simulation techniques. You will understand how you can use these techniques in HSPICE, to efficiently and accurately characterize the effect of power supply noise on a 45-nm ring oscillator as an example.

- [Webisode 9: Linear Amplifier Analysis - Predicting Gain and Noise Performance](#)

In this webisode, you will learn about gain and equivalent input noise on the example of a linear amplifier. You will also learn how to simulate and characterize related key figures of merit, such as ordered noise-contributing components, maximum gain, 3dB bandwidth, gain bandwidth product, and phase response.

- [Webisode 8: MOS Reliability Analysis - Aging Simulation and Analysis of CMOS Circuits](#)

In this webisode, you will learn about the concept of CMOS circuit aging and its simulation and analysis through HSPICE's MOS Reliability Analysis (MOSRA). You will

also learn techniques to obtain and interpret circuit degradation results for your next aging-aware design.

- [Webisode 7: MOSRA Level 3 - An Accurate and Efficient MOSFET Aging Model](#)

In this webisode, you will learn about the formulation of HSPICE's built-in MOSRA Level 3 model that accurately captures two major aging effects in MOSFETs: the hot carrier injection (HCI) effect and the bias temperature instability (BTI) effect. You will also learn what measurement data is needed to extract and validate a MOSRA Level 3 model.

- [Webisode 6: Monte Carlo-Data Mining HSPICE Analysis Results](#)

In this webisode, you will learn how to efficiently gather and correctly interpret the rich set of Monte Carlo results.

- [Webisode 5: Monte Carlo-How to Analyze Design Variations using HSPICE](#)

HSPICE provides foundry-certified device and statistical models, as well as various advanced statistical analyses and sampling methods. In this webisode, you will learn how to take advantage of these capabilities to best suit your applications.

- [Webisode 4: RUNLVL - Controlling the Accuracy and Speed of Transient Analysis](#)

This webisode will help you understand how to trade off speed and precision while maintaining HSPICE's accuracy using RUNLVL, a single option for adjusting the speed and precision of transient analysis. You will learn how to control the precision level at different time intervals of the simulation and how to reduce the precision of noncritical blocks in your circuit without reducing the overall simulation accuracy.

- [Webisode 3: IBIS-AMI - Transient and Statistical Analysis with IBIS-AMI I/O Models](#)

In this webisode, you will learn about I/O buffer models in HSPICE and how to use them in signal integrity simulations. This webisode will help you understand the benefits of the IBIS Algorithmic Modeling Interface (AMI) and how to effectively use AMI models in both statistical and full transient channel simulation. You will also learn how to mix transistor-level and IBIS-AMI models in different phases of the design.

- [Webisode 2: Eye Diagrams - Fast and Accurate Jitter Prediction for High-speed Links](#)

In this webisode, you will learn about the basic concepts of eye diagrams, inter-symbol interference and random jitter for high-speed serial and parallel channels. This webisode will help you understand how to trade off accuracy and performance at different stages of the design using HSPICE's statistical, bit-by-bit, and transient eye diagram analysis modes. You will also learn how to mix transistor-level and IBIS-AMI models in the same testbench.

- [Webisode 1: S-Element - Accurate Transient Analysis with Complex Package Models](#)

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This webisode will help you understand how to use complex package models and other S-parameter data effectively with accurate time domain simulation. You will learn how to setup and run accurate transient analysis on circuits that include S-parameter elements. You will also learn how to ensure the quality of S-parameter data using SPUTIL, the S-parameter utility in HSPICE.